

NPF GEO INCIDENTS SYSTEM OVERVIEW

Geospatial Intelligence, Biometric Authentication, & ML Crime Forecasting

Executive Summary

NPF Geo Incidents is an enterprise-grade GeoDjango web application developed for the Nigeria Police Force. The system provides law enforcement administrators and field officers with interactive spatial visualizations, secure biometrics identification, role-based regional data restrictions, and machine-learning-driven crime forecasting. By merging modern geographic databases with automated biometric pipelines, the platform enables targeted policing, efficient incident tracking, and secure record management.

Core Capabilities & Functional Modules

- 1. Interactive Heatmaps & Geospatial Visualization:** Features a full-screen, high-performance web dashboard powered by Leaflet.js and Leaflet.heat. It dynamically renders heat maps representing incident densities and plots boundaries for Nigerian States and Local Government Areas (LGAs). Officers can drill down from national scopes to specific local sectors.
- 2. Intelligent Geographic Reporting:** When submitting a security incident, the system maps exact coordinates. Using GeoDjango spatial query capabilities (SpatiaLite database backend), it automatically performs a geographic intersection to resolve and assign the corresponding Nigerian State and LGA based on the input coordinates, reducing human logging errors.
- 3. Regional Limit & Role-Based Access Control (RLAC):** Security compliance is built-in. Officer accounts are assigned a specific geographic role (National, State, or LGA). The system filters the visual map, administrative boundaries, and REST API responses based on these assignments to prevent unauthorized exposure of sensitive local files.
- 4. Suspect Biometrics & Directory:** A centralized database tracking suspect profiles including mugshots, personal attributes, and physical arrest details. Includes a simulated face-recognition search engine and fingerprint minutiae-matching utility.
- 5. Predictive Hotspots (Machine Learning):** Integrates spatial analytics modeling that projects future high-risk regions. These predictive hotspots are visualized as highlighted rings on the main map dashboard to aid in proactive patrol deployments.
- 6. Executive Analytics Dashboards:** Interactive charting using Chart.js that provides statistical summaries on incident types (theft, assault, homicide, etc.), severity indices (low, medium, high, critical), monthly crime trends, and geographical distribution of incidents.

System Architecture & Data Security

The application is built on the Django framework, utilizing robust geographical extension libraries and modern web technologies:

Component	Technology & Library	Functional Purpose
Web Framework	Django / Python	Core MVC structure, user authentication, and admin panels.

Spatial Engine	GeoDjango & SpatiaLite / GDAL	Handles geographic Point coordinates and multi-polygon state/LGA shapes.
Frontend Map	Leaflet.js & Leaflet.heat	Dynamic client-side heatmap rendering and boundary interactivity.
Data Visualization	Chart.js	Aggregates incident logs to produce monthly trendlines and bar graphs.
Security & Privacy	Fernet Database Encryption	Encrypts sensitive PII fields (reporter names, contacts, descriptions) in transit and rest.
Biometrics (ML)	Face Embedding Vector Matcher	Compares query image vectors against registered mugshot databases.
Biometrics (Finger)	Minutiae Grid Comparator	Matches ridge details, angle vectors, and coordinates of fingerprint templates.

Database Encryption & Data Integrity

To ensure compliance with data protection laws, the application integrates specialized encrypted models. Fields such as the *description*, *reported_by*, and *contact_details* in the **Incident** model inherit from custom fields (EncryptedTextField and EncryptedCharField). These fields utilize a cryptographic utility that automatically encrypts values on database writes (preventing unauthorized read access via direct SQL) and decrypts them on retrieval for authenticated officers, ensuring total confidentiality of reporters and operational details.